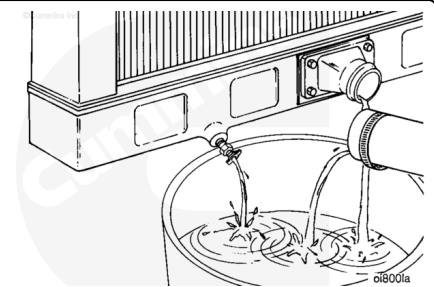


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Remove the cooling fan (if installed). Refer to Procedure 008-040 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html>)
- Remove the fan belt (if installed). Refer to Procedure 008-002 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html>)
- Drain the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html>)

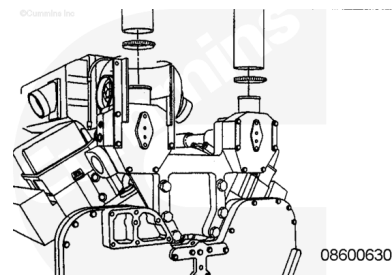
- Remove the fan hub (if installed). Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
- Remove the fan hub bracket (if installed). Refer to Procedure 008-035 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html)
- Remove the water filters (if installed). Refer to Procedure 008-006 in Section 8.
(/qs3/pubsys2/xml/en/procedures/102/102-008-006.html)
- Remove the heat exchanger (if equipped). Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-053-tr.html)

Remove

Conventional Aftercooling

Note : Shown is the K38 and K50 thermostat housing. Although different in appearance, the procedure remains the same for the QSK series engines.

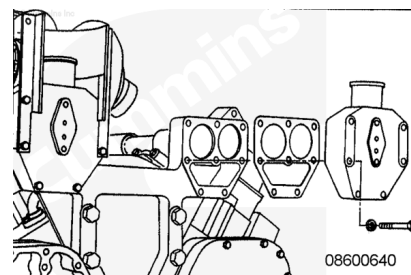
Remove both upper radiator hoses from the thermostat housing.



Remove the 12 capscrews.

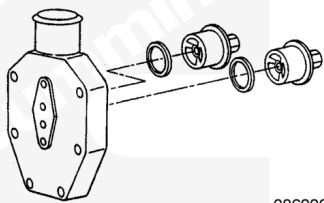
Remove the thermostat housing and gasket.

Discard the gasket.



Remove the thermostats and seals from the thermostat housing.

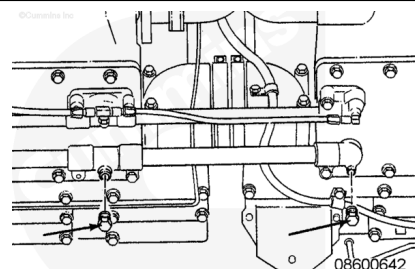
Note : Keel-cooled marine QSK50 CM2150 engines will have an insert in place of the thermostat, in the left-most position when facing the front of the engine.



08600641

LTA

Remove the pipe plug from the lower water transfer connection on aftercooler core on each bank to drain the aftercooler plumbing on engines equipped with LTA.

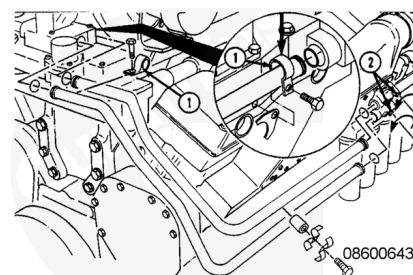


08600642

Remove the tube clamps (1) and the aftercooler plumbing on both banks.

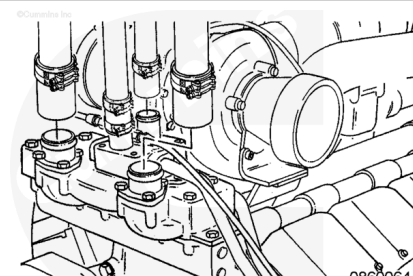
Loosen the capscrews (2) on the water transfer connection to allow flexibility.

Remove the water transfer tubes.



08600643

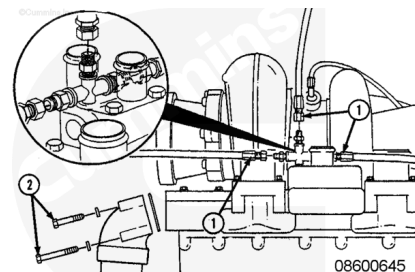
Remove the radiator hoses from the thermostat housing.



08600644

Remove the vent lines from the thermostat housing (1).

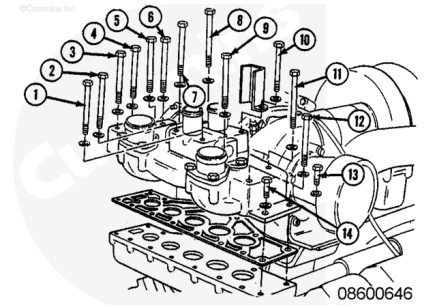
Remove the four water bypass tube capscrews (2).



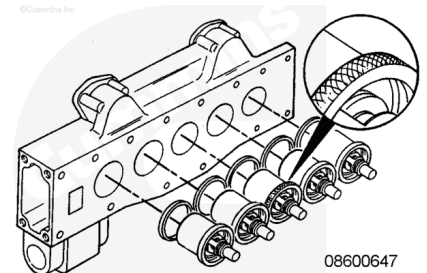
08600645

Remove the 14 capscrews and the heat shield.

Remove the thermostat housing and gasket.



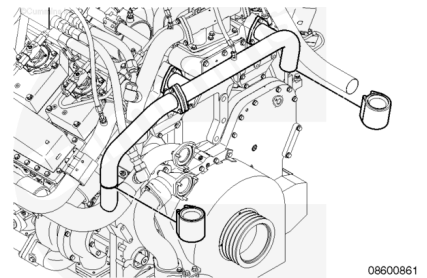
Remove the four main thermostats and the one knurled LTA thermostat in the center. The main and LTA thermostats are **not** interchangeable.



with Electronically Actuated Injector

Remove the two Straub® couplings on the thermostat outlet tube on marine engines.

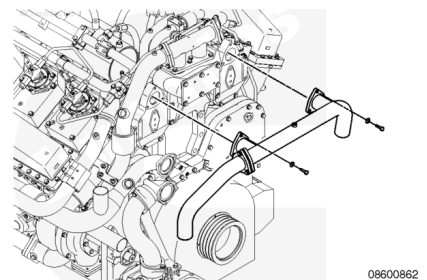
Remove the hose clamps and hoses for the industrial engines (not shown).



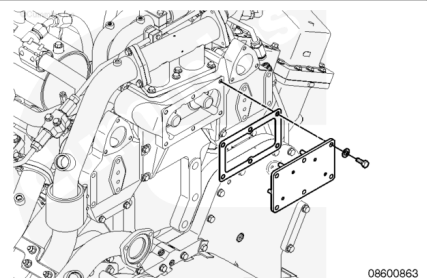
Disconnect the vent line from the thermostat outlet tube.

Remove the capscrews and washers from the thermostat outlet connections to the outlet tube.

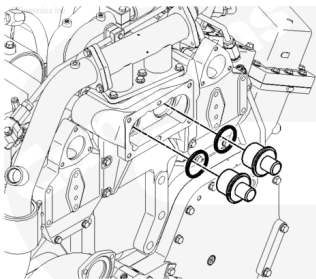
Remove the outlet tube and discard the gaskets.



Remove the six capscrews, washers, cover plate, and gaskets from the LTA thermostat housing.



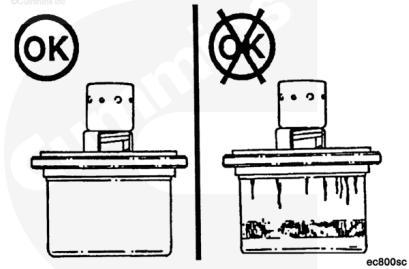
Remove the LTA thermostats and seals from the thermostat housing.



08600864

Inspect for Reuse

Check the thermostat for wear or damage. If the barrel of the thermostat is worn or fretted, it **must** be replaced.



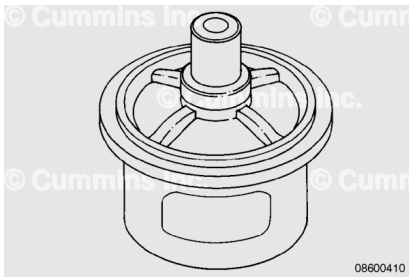
ec800sc

Test

Check the thermostat marking to see when the thermostat is designed to open.

The rated nominal temperature is marked on the side of the thermostat.

The design temperature of the thermostat is 82°C [180°F].



08600410

Engine Thermostat Initial Opening Temperature

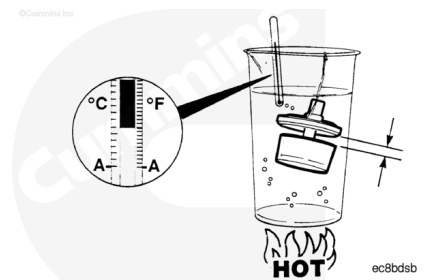
celsius		fahrenheit
82.2	MIN	180
94.4	MAX	202

LTA Thermostat Initial Opening Temperature

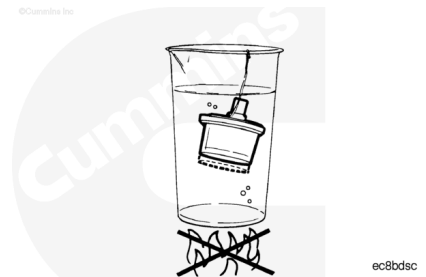
celsius		fahrenheit
73.8	MIN	165
86.1	MAX	187

Continue to apply heat until the water reaches the full open temperature for the specific thermostat being tested. The temperature is 12°C or 22°F above nominal opening temperature. At that temperature, measure the open distance of the thermostat.

The minimum open distance at full open temperature is 9.52 mm [0.375 in].



Remove the container from the heat.

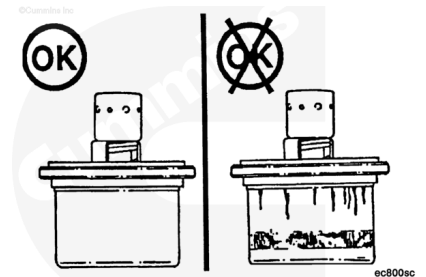


With the thermostat open, check for wear or other damage.

If the seat of the thermostat is worn or fretted, it **must** be replaced.

Check to see if the thermostat returns to the closed position.

If the barrel of the thermostat is worn or fretted, it **must** be replaced.



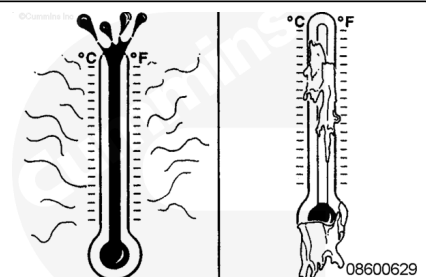
Leak Test

⚠ WARNING ⚠

Complete this procedure with engine coolant temperatures below 50°C [120°F]. Hot coolant or steam can cause serious personal injury.

This procedure has been developed to troubleshoot a thermostat failed in the open position, or experiencing excessive leakage at the seat.

This problem is often verified by observing premature warming of the thermostat housing coolant outlet pipes to the radiators. These pipes **must not** warm previous to coolant temperatures reaching the designated thermostat opening temperature.



The following procedure outlines troubleshooting a thermostat stuck in the open position:

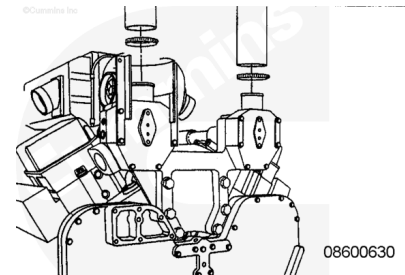
Note : All troubleshooting procedures dealing with thermostats stuck in the open position must take place with coolant temperatures below the nominal opening temperatures of the thermostats.

Two test procedures are included. One defines the procedure for the standard cooling system, the second describes engines equipped with LTA.

Conventional Aftercooling

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



Note : Shown is the K38 and K50 thermostat housing. Although different in appearance, the procedure remains the same for the QSK series engines.

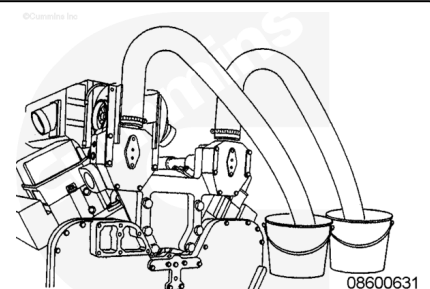
Remove the upper radiator hoses from the thermostat housing.

Coolant flowing from the thermostat housing with the hoses removed signifies a thermostat stuck in the open position. Remove the thermostat and test the operation as described in the Test section.

Install two hoses of the same size on the thermostat housing outlets long enough to reach a remote dry container.

Install and tighten the hose clamps on the housing outlets.

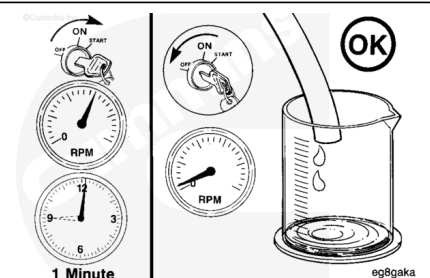
Place the end of the hoses in two dry containers.



Operate the engine at rated rpm for 1 minute.

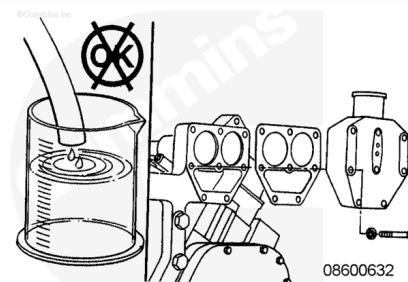
Shut the engine off, and measure the amount of coolant collected in each container.

The amount of coolant collected in either container **must not** exceed 200 cc [7 fluid oz].



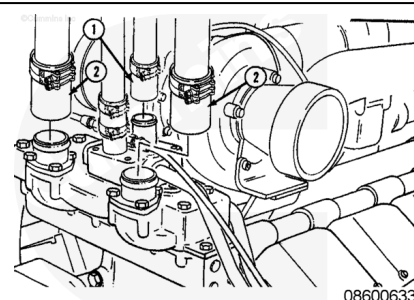
If more than 200 cc [7 fluid oz] of coolant is collected from either container, the thermostat or thermostat seal is leaking.

Remove the thermostat and test operation as described in the Test section.

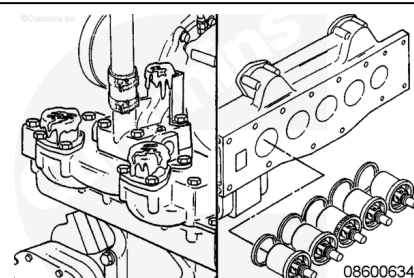


LTA

Remove the LTA radiator inlet hose (1) and the two main engine coolant outlet hoses (2) to the radiator.

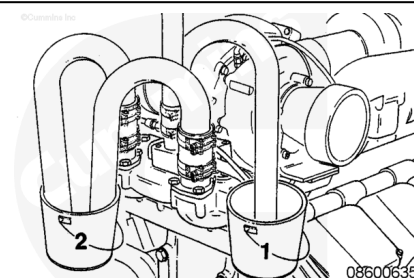


Coolant flow from the thermostat housing indicates a thermostat is stuck in the open position. Replace the thermostats.



Install a hose from the LTA Thermostat housing to a remote container (1). Install a hose from each of the main engine thermostat coolant outlets to a common remote container (2).

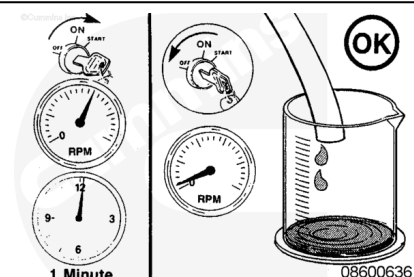
Partially fill each hose with coolant. Place the end of each hose in its respective container. Allow the hoses to drain in this position. Remove the coolant from the containers.



Operate the engine at rated rpm for 1 minute.

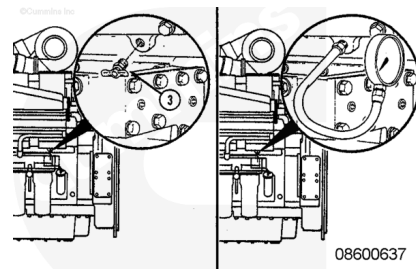
Turn off the engine and measure the amount of coolant in the container.

The amount of coolant collected from the main engine thermostats **must not** exceed 400 cc [14 fluid oz].

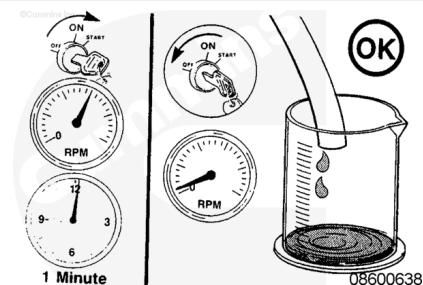


Remove the draincock (3). Use a gauge of 420 kPa [60 psi] minimum capacity.

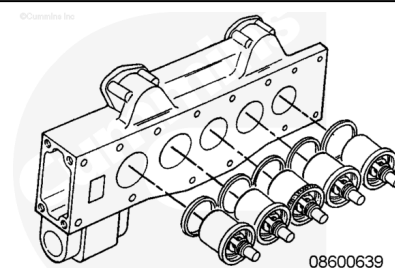
Remove any coolant from the container collecting the LTA thermostat leakage.



Start the engine. Note the block coolant pressure. Increase the engine speed until the coolant pressure reaches 170 kPa [25 psi]. Maintain this engine speed for 1 minute. Shut the engine off and measure the amount of coolant collected in the container.



The amount of coolant collected from the LTA thermostat **must not** exceed 100 cc [3.5 fluid oz]. If more is collected, the thermostat or thermostat seal is leaking. Remove the thermostat(s) and test.



Install

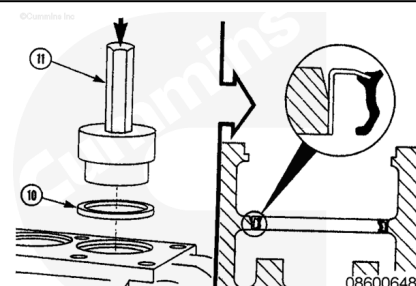
Conventional Aftercooling

Note : Shown is the K38 and K50 thermostat housing. Although different in appearance, the procedure remains the same for the QSK series engines.

The seal **must** be installed with the part number up.

Use a mallet and seal driver, Part Number 2892254, to install the seal.

Install the seal no more than 0.51 mm [0.020 in] below the top of the cast edge.

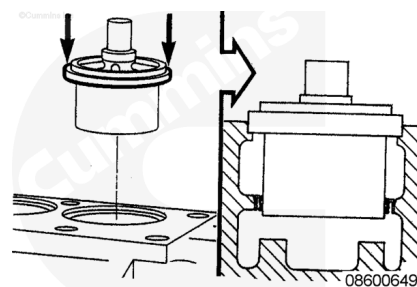


Install the thermostat by pushing on the outer rim.

Note : If the engine being serviced is a K50, be sure the thermostat is installed.

The K50 thermostats contain a nitrile rubber seat vulcanized to the brass collar at the barrel seat. This seat prevents coolant leakage and wear. The thermostat with the rubber seat **must** be on K50 engines. The thermostat with the rubber seat is **not** required for the K38 engine, but can be installed.

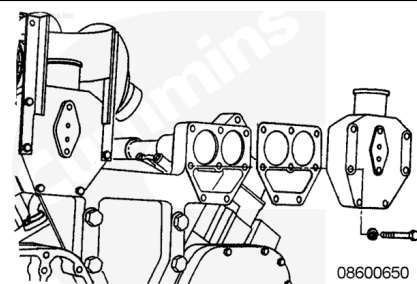
Note : Keel cooled marine QSK50 CM2150 engines will have an insert in place of the thermostat, in the left most position when facing the front of the engine.



Install the thermostat housing.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

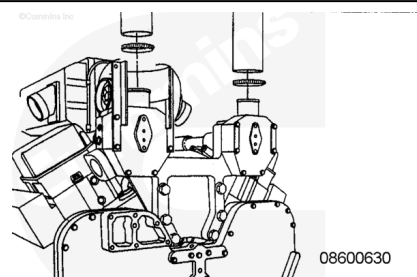


On K38 and K50 engines, install the two upper radiator hoses.

Tighten the hose clamps.

Torque Value: 6 n•m [53 in-lb]

On the QSK38 and QSK50 engines, connect the water lines to the triangle, three bolt flanges, using new gaskets.

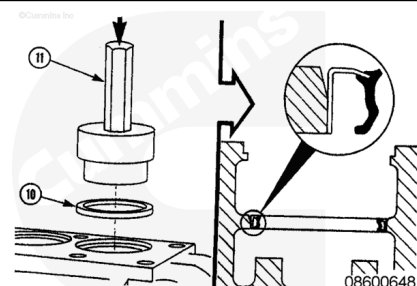


LTA

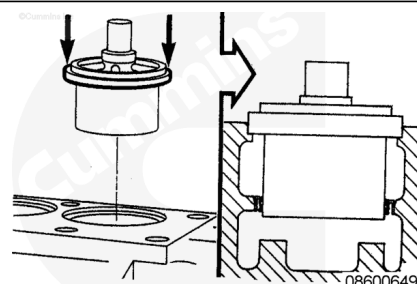
The seal **must** be installed with the part number up.

Use a mallet and seal driver, Part Number 2892254, to install the seal.

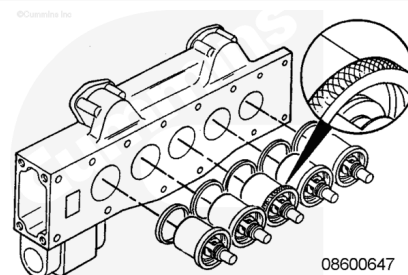
Install the seal no more than 0.51 mm [0.020 in] below the top of the cast edge.



Install the thermostat by pushing on the outer rim.



Install the four main engine thermostats and the one knurled LTA thermostat.



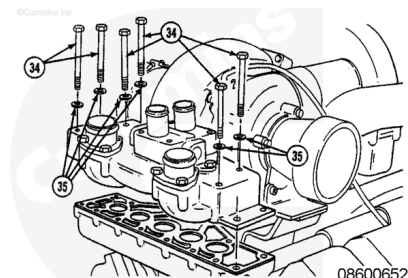
Install the thermostat housing.

Torque Value: 45 n•m [33 ft-lb]

Install the radiator hoses at the thermostat housing.

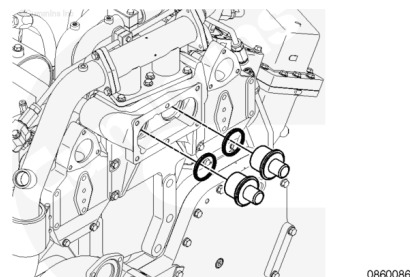
Tighten the hose clamps.

Torque Value: 6 n•m [50 in-lb]



with Electronically Actuated Injector

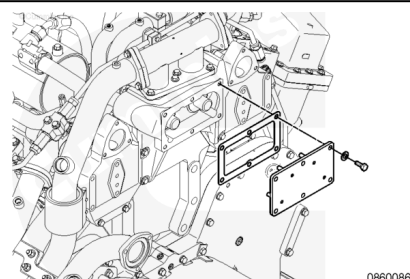
Install the LTA thermostat seals and thermostats into the thermostat housings.



Install the new gasket, thermostat cover, washers, and capscrews into the LTA thermostat housing.

Tighten the capscrews

Torque Value: 45 n•m [33 ft-lb]

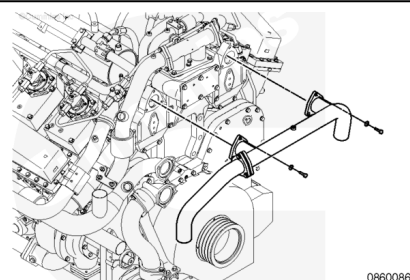


Install new gaskets and the thermostat outlet tube, washers, and capscrews to the outlet connection of the LTA thermostat housing.

Tighten the capscrews

Torque Value: 45 n•m [33 ft-lb]

Install the vent line to the thermostat outlet tube.



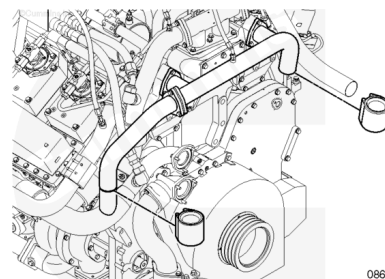
Install the two Straub® couplings onto the thermostat outlet tube on marine engines.

Torque Value: 28 n•m [21 ft-lb]

Install the hose and two hose clamps onto the thermostat outlet tubes on industrial engines (not shown).

Tighten the hose clamps.

Torque Value: 6 n•m [50 in-lb]

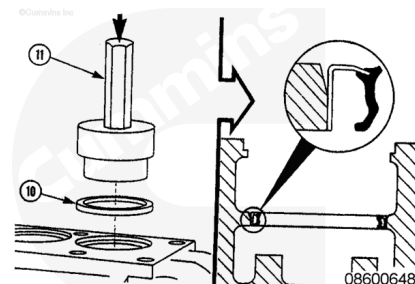


Gas Compression

Note : The seal must be installed with the part number up.

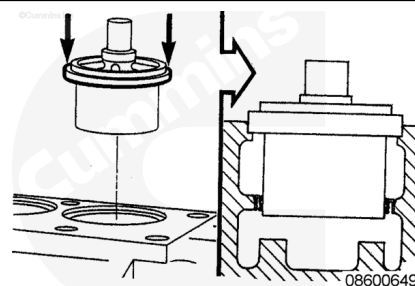
Use a mallet and seal driver, Part Number 2892254, to install the seal.

Install the seal no more than 0.51 mm [0.02 in] below the top of the cast edge.



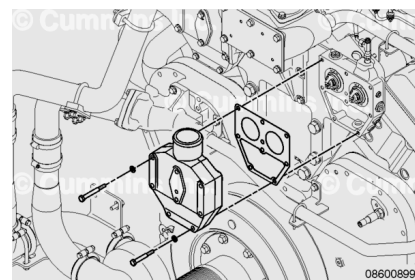
Install the thermostat by pushing on the outer rim.

The K50 thermostats contain a nitrile rubber seat vulcanized to the brass collar at the barrel seat. This seat prevents coolant leakage and wear. The thermostat with the rubber seat **must** be on K50 engines. The thermostat with the rubber seat is **not** required for the KTA38GC engine, but can be installed.



Install the thermostat housing.

Torque Value: 40 n•m [30 ft-lb]



Finishing Steps

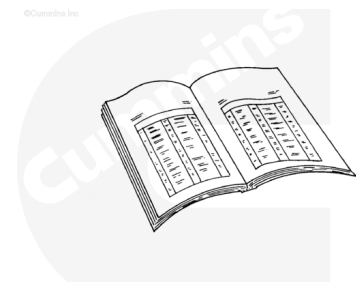
⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the fan hub bracket (if installed). Refer to Procedure 008-035 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-035-tr.html)
- Install the fan hub (if installed). Refer to Procedure 008-036 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-036-tr.html)
- Install the fan belt (if installed). Refer to Procedure 008-002 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)
- Install the cooling fan (if installed). Refer to Procedure 008-040 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-040-tr.html)
- Install the heat exchanger (if equipped). Refer to Procedure 008-053 in Section 8.
(/qs3/pubsys2/xml/en/procedures/28/28-008-053-tr.html)
- Install the water filters (if installed). Refer to Procedure 008-006 in Section 8.
(/qs3/pubsys2/xml/en/procedures/102/102-008-006.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (/qs3/pubsys2/xml/en/procedures/28/28-008-018-tr.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)
- Operate the engine until the coolant temperature reaches 70°C [158°F], and then check for leaks.



ck800wa

Last Modified: 24-Sep-2015
